A Cryogenic Helium Gas Target for Direct Reactions with Radioactive Ion Beams

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Goal 2: Obtain a more accurate experimental and theoretical understanding of exotic and stable nuclei below and above decay thresholds

MOTIVATION

- ^{3}He-induced reactions are a useful probe of the structure of atomic nuclei
  - Reactions like (^{3}He,*p*), (^{3}He,*n*), and (^{3}He,*d*) populate excited states in nuclei, including unstable nuclei using radioactive ion beams
- Deuteron-transfer reactions such as (^{3}He,*p*) allow for investigation into A=9, T=1/2 mirror nuclei ^{9}Be and ^{9}B using stable ^{7}Li and unstable ^{7}Be beams, respectively

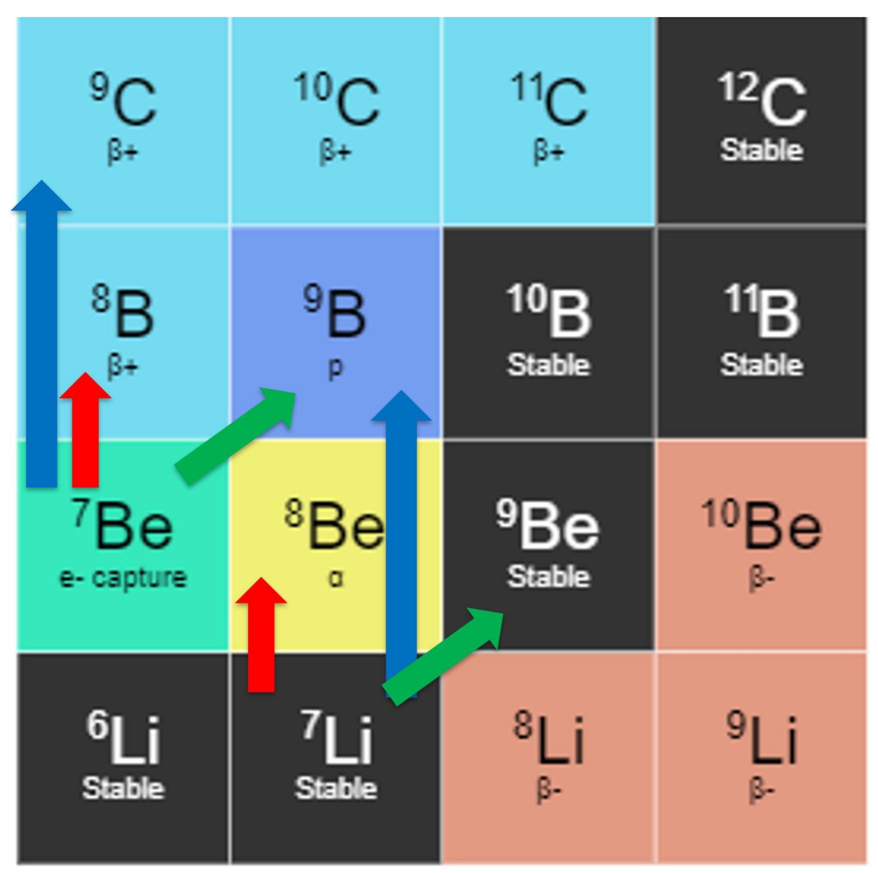

Figure 1: (^{3}He,p) deuteron transfer reactions are shown with green arrows. (^{3}He,d) proton transfer reactions are shown with red arrows, and (^{3}He,n) double-proton transfer reactions are shown with blue arrows.

- ^{9}B has been studied via theoretical and experimental means for decades with no significant agreement between measured and predicted energies and widths of the first-excited state
- Conversely, ^{9}Be has been extensively studied and understood, including the mirror first excited state [1,2] (E = 1.68 MeV, Γ = 214 keV, $J^\pi = \frac{1}{2}^+$) [3]
- The mirror first-excited states in ^{9}Be and ^{9}B can be populated via (^{3}He,*p*) reactions

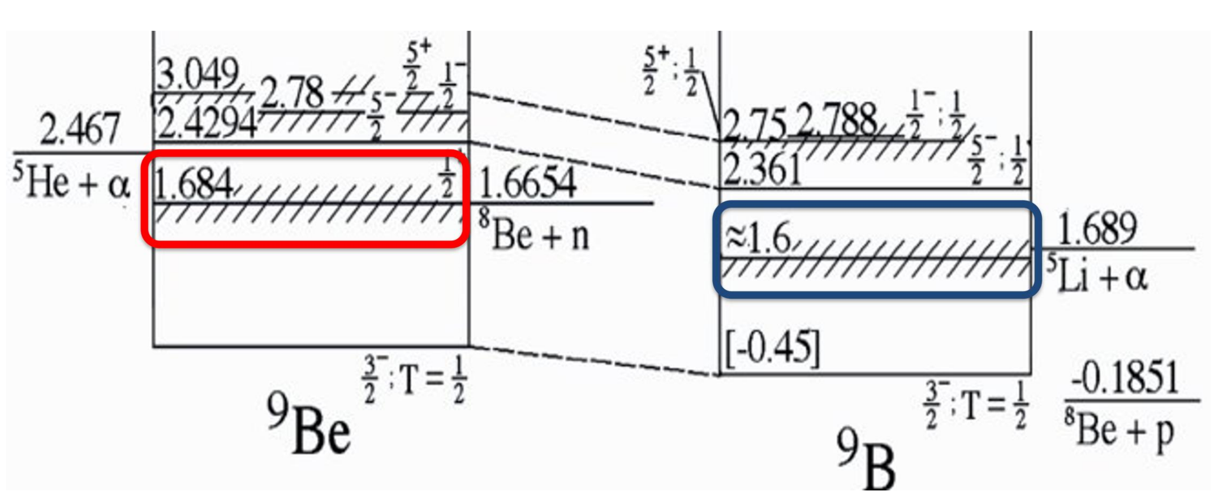

Figure 2: First excited states in mirror nuclei ^{9}Be (red) and ^{9}B (blue).

- Additionally, ^{9}C is an isotope in which all excited states are unstable to single- and double-proton emission
  - Mirror reaction ^{7}Li(*t*,*p*) used to populate neutron-unbound excited states in the mirror nucleus ^{9}Li [4,5]

LSU Gas Target 1.0

- LSU gas target modeled after HELIOS [6] gas target
- Target body is a disk with window flanges on either side
  - Window material: 1-mg/cm^2 Kapton (7.5-µm)
  - Aperture diameter: 0.25-0.5"
  - Target linear thickness variable via flanges with different reentrant windows
  - Titanium body with internal cooling channel

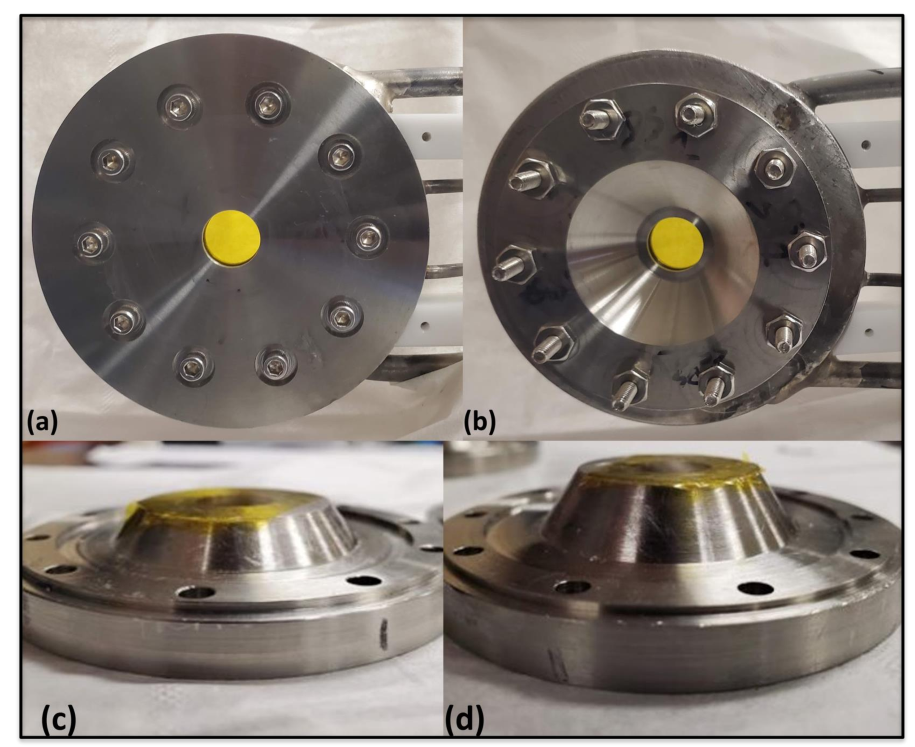

Figure 3: A view of the upstream (a) and downstream (b) flanges of the LSU gas target before commissioning at the TAMU CI. The downstream flange can be designed to limit the linear thickness to 5mm (c) or 3mm (d).

- Successfully commissioned at Texas A&M University Cyclotron Institute using the MARS beamline
  - Gas at room temperature
  - Average ^{3}He target thickness ~40-µg/cm^2 (~400 Torr average)
  - 10^9 ^{7}Li ions/second for hours (no window failures)

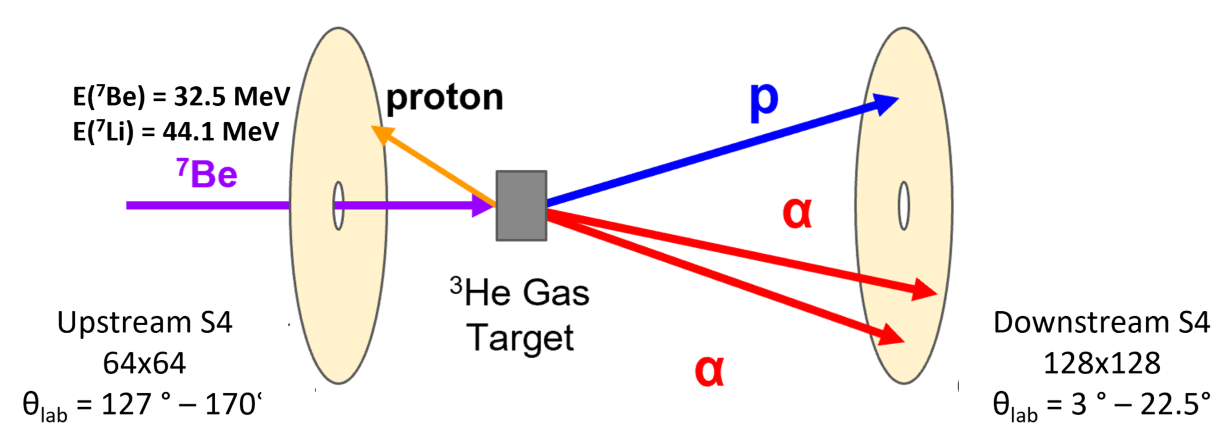

Figure 4: Cartoon of the detector and target setup. Relevant beam energies and lab angle coverages are shown.

- Evidence for ^{3}He-induced reactions
  - ^{3}He(^{7}Li,*d*)^{8}Be easily seen above background in gas-in target runs and disappears in gas-out target runs

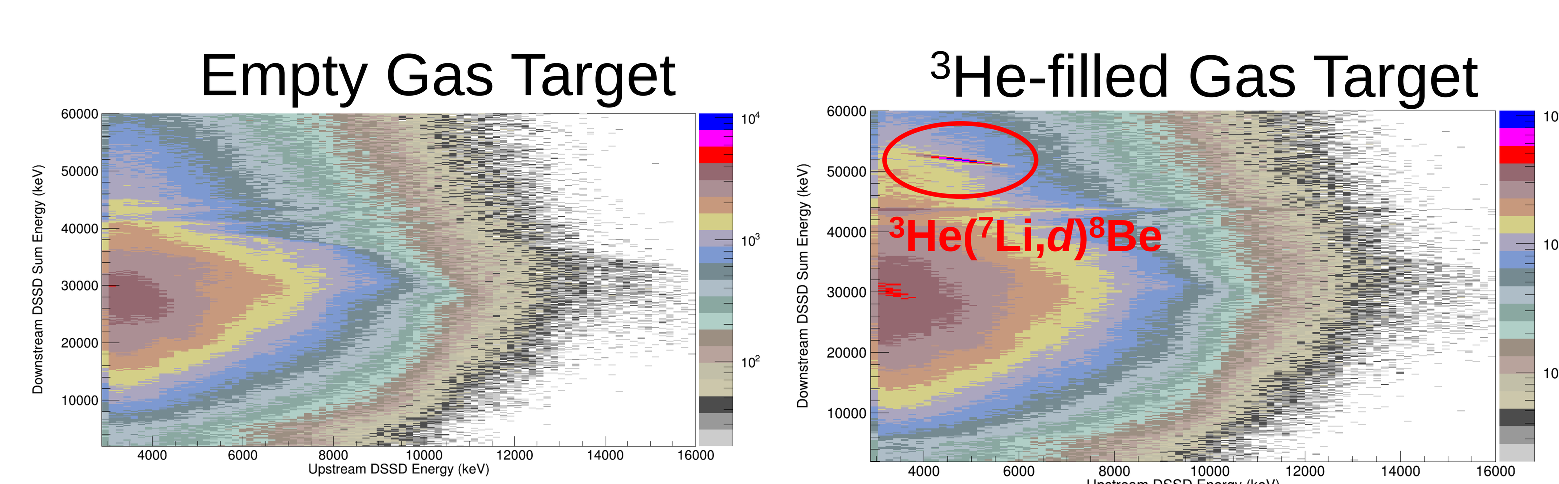

Figure 5: Downstream sum energy versus upstream energy for empty gas target runs (left) and ^{3}He-filled gas target runs (right).

LSU Gas Target 2.0

- A new version of the LSU Gas Target is in development to simplify the process of assembling and operating the target
  - Stainless steel body is simpler to machine and can be brazed/soldered
  - Minimize body size and thus thermal mass
  - VCR connectors to minimize gas and LN$_2$ losses
  - Mounting face identical to original target, allows for reuse of currently existing window flanges
  - Features a ¼"-OD copper cooling line soldered into a semi-circular groove on lateral surface of the target body

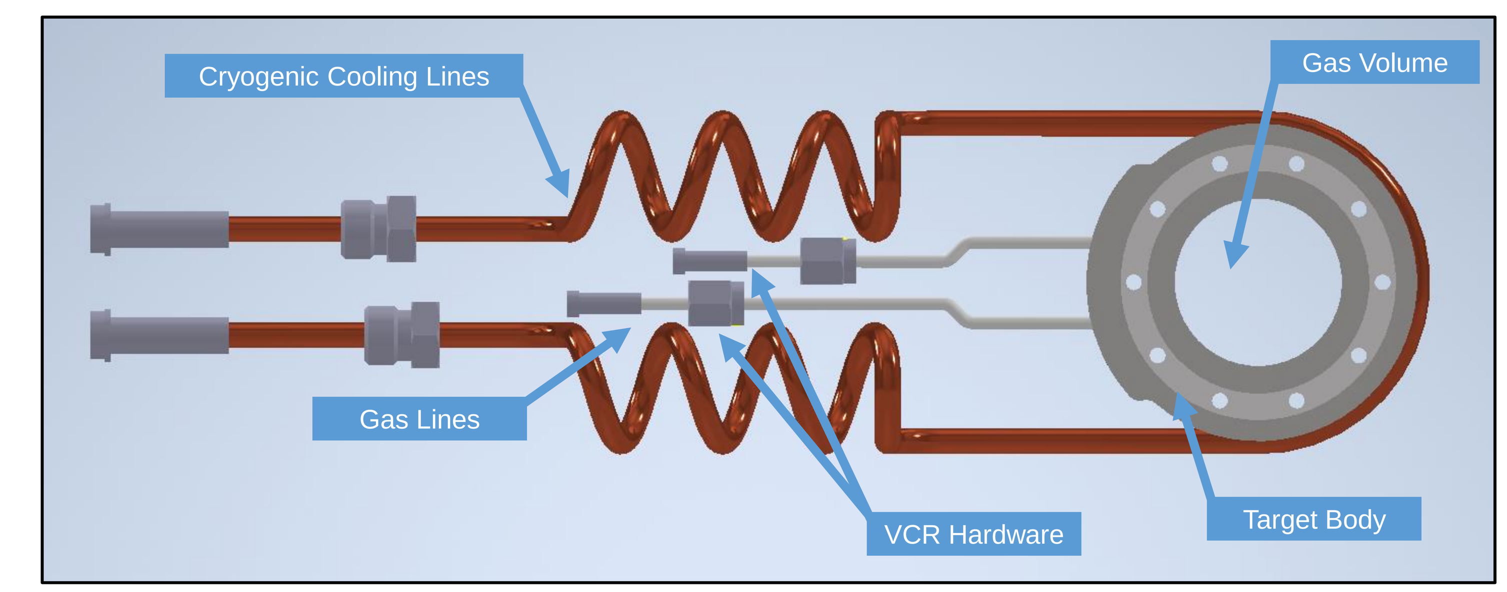

Figure 6: Rendering of the LSU Gas Target 2.0 body with cooling lines. Coiled cooling lines allow for stress reduction on the tube joints during repeated thermal cycling. VCR connectors will be used to connect the coils to the gas target body and the gas LN$_2$ vacuum feedthrough.

- Target body machined at LSU, beginning process of soldering cooling and gas lines
- Testing at LSU will include:
  - Cryogenic cycling and cooling rate
  - Pressure cycling
  - Cryogenic pressure testing
- After successful testing, the target will be commissioned at the Texas A&M University Cyclotron Institute using the MARS beamline
  - ^{3}He(^{7}Be,p)^{9}B in 2025
  - Future measurement ^{3}He(^{7}Be,*n*)^{9}C at MARS

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References

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